

# NINAAD ADHVARYU

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🐙 [github.com/adhvaryu-athena](https://github.com/adhvaryu-athena) 🐙 [github.com/Ninaad-Adhvaryu](https://github.com/Ninaad-Adhvaryu) 🌐 [ninaad-adhvaryu.github.io](https://ninaad-adhvaryu.github.io)

## Introduction

I am a computational physicist and data scientist who builds analytical pipelines and AI tooling for problems that resist clean categorisation. My background is in Python, numerical methods, and machine learning; recent work at Athena Education extends that into LLM systems used daily by colleagues, an internal submission tool now on PyPI, and the prototype of a platform productionised by the company's engineering team. Across academic and applied threads I care about making technical work legible to collaborators, students, and decision-makers.

## Technical Skills

**AI / LLMs:** Anthropic and OpenAI APIs, Claude Projects (Enterprise admin), prompt engineering, RAG patterns, structured-output JSON, LLM-as-judge; Hugging Face Transformers including Whisper, MarianMT, FinBERT, SciBERT

**ML / Data:** NumPy, pandas, SciPy, scikit-learn, PyTorch, CatBoost, pyGAM, statsmodels; supervised learning, time-series modelling, cross-validation, evaluation (ROC/PR, Brier)

**Integrations:** Gmail, Google Drive, Google Calendar, Notion (API + MCP), Airtable, Zoom Server-to-Server OAuth; FastAPI, Streamlit

**Geospatial:** QGIS, GeoPandas, rasterio, osmnx; raster/vector processing, CRS handling

**Languages & Workflow:** Python (proficient), MATLAB, Bash, Git, GitHub, Linux, LaTeX, Jupyter/Colab

## Current Role

### Research Mentor

Athena Education

Jul 2025 – Present

Gurgaon / New Delhi, India

- Mentored **66 student research projects** in 10.2 months; **93.8% HF1/HF2 quality rating against a 60% organisational target**; **13 papers accepted** across 6 venues (SIGMA XI ×7, IRJET ×2, NHSJS, CREST Gold, Curieux, IC<sup>2</sup>UHI) with **23 further papers under review**.
- Built and shipped **nhsjs-tools** 🐙 (Streamlit + `pip install nhsjs-tools`), encoding journal submission rules as code; cut per-paper formatting from ~2.5 hours to ~30 minutes and adopted by other mentors in the team.
- Designed the **Pangea Toolkit** 🐙 (v1 → v6, with v5.2 retired on usage data) – a research-ideation system grounded in a corpus of 13 NHSJS reviewer PDFs and 685 accepted papers, loaded as Project Knowledge into per-student Claude Projects; now the default ideation workflow for the research-mentor team.
- Co-architected and vibe-coded the prototype of **Athena Intelligence** (~5,700 LOC; Python/FastAPI backend, vanilla JS frontend; integrating Zoom, Google Drive, Calendar, Airtable with Anthropic Haiku/Sonnet routing); demoed to leadership, handed off to the product and IT teams for productionisation, and now used company-wide.
- Promoted from Associate Manager to Manager within 10 months.

## Selected Technical Projects

*Projects developed across my role as a Research Mentor at Athena Education (2025 – Present).*

### YouTube Finance Video → Ticker-Level Sentiment | LLM/NLP, FinBERT, Ollama 🐙

- Built a pipeline that turns YouTube finance videos into per-ticker sentiment and buy/sell/hold signals via transcript extraction, regex-based ticker canonicalisation, and aspect-based classification with FinBERT plus zero-shot MNLi baselines.
- Layered a structured JSON-only local LLM (Ollama) on top for aspect-level labelling; output is a time-stamped, per-ticker CSV with daily aggregation tables.

### Endangered-Language Speech-to-Text & Translation | Whisper, MarianMT 🐙

- Built an ASR + translation pipeline for a low-resource language using Whisper for speech-to-text and MarianMT-style translation models on Mozilla Common Voice-style datasets.
- Integrated audio preprocessing, NLP, and benchmarking with WER and BLEU into a reproducible audio → transcript → translation pipeline with comparative performance visualisations.

### Exoplanet Transit Classification | Python, CatBoost, TSFresh

- End-to-end pipeline to classify exoplanet transit candidates from Kepler/TESS light curves using TSFresh-style feature extraction and CatBoost, benchmarked against published LightGBM baselines.
- Implemented the ML workflow: data cleaning, time-series preprocessing, cross-validation, hyperparameter tuning, and evaluation with ROC/PR curves and confusion matrices in reproducible notebooks.

### Hawkeye Cricket Analytics – Next-Ball Wicket Prediction | PyTorch, pyGAM 🐙

- Built a ball-tracking analytics pipeline on Men's IPL Hawkeye data (~140,000 deliveries) to forecast next-ball wicket probability and learn situation-conditioned wicket surfaces over line/length.
- Engineered match-context features and trained PyTorch GRU sequence models and logistic GAMs (pyGAM) with GroupKFold CV and PR-AUC/Brier evaluation, deriving prescriptive line-length maps and lift@top-decile metrics.

## Mangrove Cover Change – Mumbai Coastline | GeoPandas, rasterio

- Built a geospatial workflow comparing Global Forest Watch and Global Mangrove Watch datasets (2000–2020) for mangrove cover change along Mumbai’s coastline, including projection/resolution harmonisation across EPSG:32643 and the GMW mask.
- Computed temporal loss series, spatial agreement metrics (Jaccard, precision/recall) over a 1,509 ha mangrove extent, and generated hotspot maps and comparison tables for environmental planning use.

## Wi-Fi Drone Placement in Disaster Zones | Python

- Designed a simulator to optimise placement of Wi-Fi drones over 3D terrain using DEM/heightmaps, line-of-sight ray tracing, and RF path-loss models to maximise coverage under physical constraints.
- Implemented heuristic strategies (greedy set-cover, centroid partitioning) with parameter sweeps over drone count, signal radius, and terrain types, analysing coverage heatmaps and coverage-vs- $k$  curves.

## Education

### University of Stuttgart

*International Physics Masters, GPA: 2.3*<sup>1</sup>

**2020 – 2023**

*Stuttgart, Germany*

### Shiv Nadar University

*BSc (Research) Physics with minor in Mathematics, GPA: 7.51/10*

**2015 – 2019**

*New Delhi, India*

- Undergraduate thesis: *Developing Calculations for X-ray Absorption Spectra near the Diffraction Limit* (Advisor: Prof. Dr. Sen Gupta).

## Research Experience

### Dipolar Quantum Gases under Rotation

*Advisor: Prof. Dr. Tilman Pfau (PI5)*

**2021 – 2023**

*University of Stuttgart*

- Developed and tested two numerical solvers (preconditioned Conjugate Gradient and Imaginary Time Evolution with Strang operator splitting) for the extended, rotating Gross-Pitaevskii equation, including the Lee-Huang-Yang quantum-fluctuation correction, on a rotating dipolar BEC of  $\sim 75,000$  dysprosium ( $^{164}\text{Dy}$ ) atoms on a  $128^3$  spatial grid.
- Introduced a Cascadic Multigrid scheme that reduced computation time by **5–22 $\times$**  by seeding fine-grid runs from coarse solutions; cross-validated ground states across both solvers and recovered hexagonal Abrikosov-like vortex lattices.
- Characterised convergence, stability, and spatial-resolution effects on ground-state energies; automated parameter sweeps and diagnostic plotting for reproducible analysis. [\[PDF\]](#) [\[Write-up\]](#)

## Teaching & Research Experience

### Teaching Assistant (Instructor Role) | *Emergence: Complex Systems in Science & Society, Ahmedabad University*

**Apr 2025**

- Designed and delivered a four-week seminar on complexity across physics, biology, and social systems, integrating quantitative models with qualitative readings.
- Guided students from topic proposals to 20-page interdisciplinary research papers, with emphasis on clarity of modelling assumptions and evidence.

### Teaching Assistant | *Feminism is for Everybody, Ahmedabad University*

**Mar 2025**

- Supported an interdisciplinary course on feminist theory and practice, facilitating discussions and helping students connect abstract concepts to contemporary social issues.
- Provided feedback on written assignments and projects, focusing on argumentation, clarity, and engagement with readings.

### Teaching Assistant | *Foundation Programme, Ahmedabad University*

**Feb 2024 – Mar 2025**

- Designed and facilitated interdisciplinary, project-based learning across studios including Environment & Climate Change, Water, and Democracy & Justice.
- Co-developed the Foundation Programme Handbook to consolidate lesson plans, activities, and onboarding material; trained Teaching Assistants for the Monsoon 2024 semester.
- Assessed student work with structured rubrics and provided detailed, iterative feedback on research design, argumentation, and written/oral communication.

### Research Intern | *Center for Advanced Research in Building Science & Energy, CEPT University*

**Jun 2019 – Feb 2020**

- Supported building-energy and indoor-environment research, adapting theoretical mould-growth models to Indian climatic conditions and assisting with data collection and analysis.

### Summer Intern | *interfacial Science and Surface Engineering Lab, University of Alberta*

**May – Aug 2018**

- Conducted experiments on sessile droplet evaporation and surface wettability, contributing to the design and interpretation of lab measurements on droplet dynamics.

<sup>1</sup>In the German GPA system, 1 is the best grade (excellent), and 6 is the lowest (insufficient). My grade equivalent in US GPA is 2.7/4.